

Integration Wrap-up

Today's Goals

- ◆ You should be able to employ different strategies to evaluate integrals.
- ◆ You should be able to recognize when an integral will be difficult or impossible to evaluate exactly.
- ◆ You should be able to find formulas for functions when given information about their derivatives.

1. Evaluate each of the following integrals. You won't be able to evaluate one of them—which one is it?

(a) $\int \sin(\sin x) \cos x \, dx$

(b) $\int_e^{e^e} \frac{(\ln x)^5}{x} \, dx$

(c) $\int_e^{e^e} \frac{\ln(x^5)}{x} \, dx$

(d) $\int_0^1 \arctan x \, dx$

(e) $\int_0^2 e^{-x^2} dx$

(f) $\int_{-1}^1 x^3 e^{-x^2} dx$

(g) $\int \cos \sqrt{x} dx$

(h) $\int_0^1 \frac{2x+3}{x^2+1} dx$

2. Approximate the integral you weren't able to evaluate in [#1](#). (Use a calculator to obtain a decimal result.)
Can you determine the maximum error in your approximation?

3. A rocket starts on the ground at rest. At a time of t seconds after it is launched, its acceleration is

$$a(t) = 14e^{t/100} - 10,$$

measured in (meters/second)/second.

- (a) The rocket has a velocity of 0 meters/second at the moment it is launched. Compute the velocity of the rocket t seconds after it is launched.

- (b) How far does the rocket travel in the first 30 seconds of its movement?